

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2024 P1HR Worked Solutions

May 2024 | P1HR

27 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Statistics Toolkit**

May 2024 P1HR - May 2024 | P1HR

1 Here are six cards.

Five of the cards have a number written on them.

16	15	3	2	9	
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Work out the number that should be written on the last card so that the mean of the six numbers will be 11

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

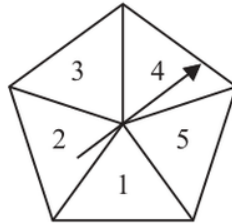
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

2 Here is a biased spinner.

The table gives information about the probability that, when the spinner is spun once, it will land on each number.

Number	1	2	3	4	5
Probability	$2x$	0.27	0.04	x	0.12

Alexis is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on an odd number.

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2

Marks: 4

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 5****TOPIC: Percentages**

May 2024 P1HR - May 2024 | P1HR

- 3** Norberto sells white loaves of bread and brown loaves of bread.

He sells a total of 200 loaves such that

the number of white loaves sold : the number of brown loaves sold = 3 : 2

Norberto sells the white loaves for £1.50 each.

He sells the brown loaves for £1.75 each.

40% of the price of a white loaf is profit.

60% of the price of a brown loaf is profit.

Work out Norberto's total profit when he sells all 200 loaves.

£.....

(Total for Question 3 is 5 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 5

TOPIC: Percentages

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 3**TOPIC: **Algebraic Proof**

May 2024 P1HR - May 2024 | P1HR

4 Show that $2\frac{1}{3} \div 5\frac{1}{4} = \frac{4}{9}$

(Total for Question 4 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 4

Marks: 3

TOPIC: **Algebraic Proof**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2024 P1HR - May 2024 | P1HR

- 5** Slavomir invests 5200 euros in a savings account for 4 years.
He gets 2.5% per year compound interest.

Work out how much money Slavomir will have in the savings account
at the end of 4 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 5 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 5

Marks: 3

TOPIC: **Compound Interest & Depreciation**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Standard & Compound Units**

May 2024 P1HR - May 2024 | P1HR

- 6 The diagram shows a solid wooden cylinder.

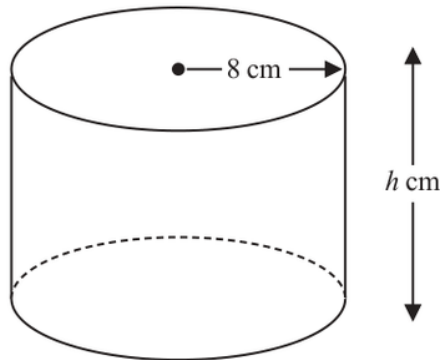


Diagram **NOT**
accurately drawn

The cylinder has radius 8 cm and height h cm.
The volume of the cylinder is 1208 cm^3

- (a) Work out the value of h
Give your answer correct to the nearest whole number.

$h = \dots\dots\dots$
(2)

The density of the wood is 1.25 g/cm^3

- (b) Work out the mass of the cylinder.
Give your answer in kilograms.

$\dots\dots\dots$ kilograms
(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6

Marks: 4

TOPIC: **Standard & Compound Units**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 7****Marks: 9****TOPIC: Factorising**

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7 (a) Simplify $g^9 \div g^2$
(1)(b) Expand $5k^2(k^3 + 4)$
(2)(c) (i) Factorise $x^2 - 2x - 63$
(2)(ii) Hence, solve $x^2 - 2x - 63 = 0$
(1)(d) Solve the inequality $7 - 2y < 3y - 12$
(3)**(Total for Question 7 is 9 marks)**

PAST PAPER SOLUTION

Q#: 7

Marks: 9

TOPIC: **Factorising**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 6****TOPIC: Area & Perimeter**

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- 8 The diagram shows a trapezium, $ABCD$

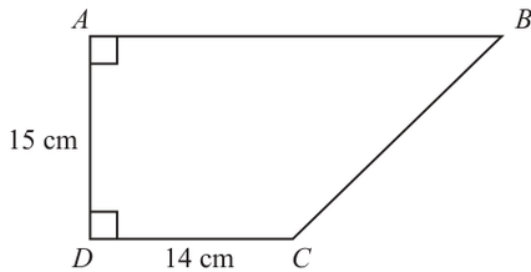


Diagram **NOT**
accurately drawn

DAB and ADC are right angles.

$$AD = 15 \text{ cm} \quad DC = 14 \text{ cm}$$

The area of the trapezium is 360 cm^2

Work out the perimeter of the trapezium.

..... cm

(Total for Question 8 is 6 marks)

PAST PAPER SOLUTION

Q#: 8

Marks: 6

TOPIC: **Area & Perimeter**

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WORKED SOLUTION

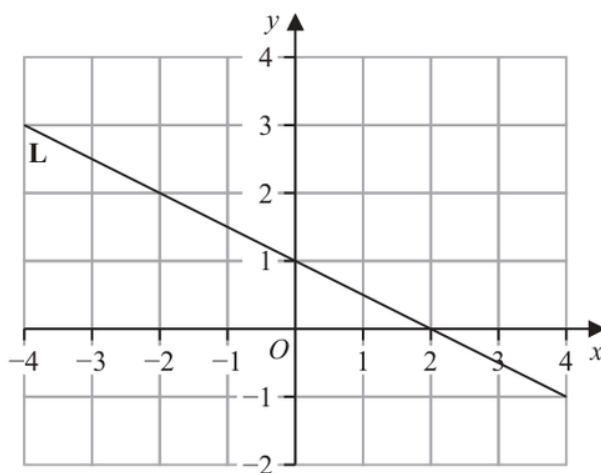
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Linear Graphs ($y = mx + c$)**

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9 Line **L** is drawn on the grid.Find an equation for **L**Give your answer in the form $y = mx + c$

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Linear Graphs** ($y = mx + c$)

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 2**TOPIC: **Statistics Toolkit**

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10 Here are the numbers of goals scored by a hockey team in its 11 games this season.

0 1 2 2 3 4 4 6 7 9 11

Work out the interquartile range of the numbers of goals.

(Total for Question 10 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 10

Marks: 2

TOPIC: **Statistics Toolkit**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

May 2024 P1HR - May 2024 | P1HR

11 $A = 2^5 \times 5 \times 7^2$

$$B = 2^3 \times 5^3 \times 7^4$$

- (a) Write down the highest common factor (HCF) of $5A$ and $2B$
Give your answer as a product of prime factors.

.....
(2)

$$A = 2^5 \times 5 \times 7^2$$

$$B = 2^3 \times 5^3 \times 7^4$$

- (b) Work out the value of $(AB)^2$
Give your answer as a product of prime factors.

.....
(2)

.....
(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

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12 Solve the simultaneous equations

$$4x + 3y = 9.6$$

$$6x + 5y = 16.8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 12

Marks: 4

TOPIC: **Simultaneous Equations**

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WORKED SOLUTION

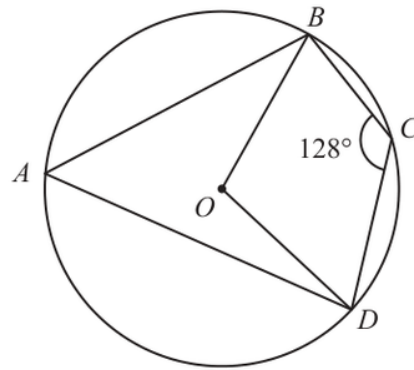
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 5****TOPIC: Circle Theorems**

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13Diagram **NOT**
accurately drawn A , B , C and D are points on a circle, centre O Angle $BCD = 128^\circ$ Work out the size of angle OBD

Give a reason for each stage of your working.

angle $OBD = \dots\dots\dots^\circ$ **(Total for Question 13 is 5 marks)**

PAST PAPER SOLUTION

Q#: 13

Marks: 5

TOPIC: **Circle Theorems**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 6****TOPIC: Expanding Brackets**

May 2024 P1HR - May 2024 | P1HR

14 (a) Expand and simplify $(3x + 1)(2 - x)(4 + x)$
(3)(b) Simplify fully $\left(\frac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$
(3)

(Total for Question 14 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 6

TOPIC: **Expanding Brackets**

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WORKED SOLUTION

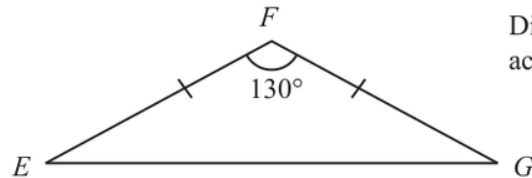
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

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15 The diagram shows isosceles triangle EFG Diagram **NOT**
accurately drawn

$$EF = GF$$

$$\text{Angle } EFG = 130^\circ$$

The area of triangle EFG is 74 cm^2 Work out the length of EF

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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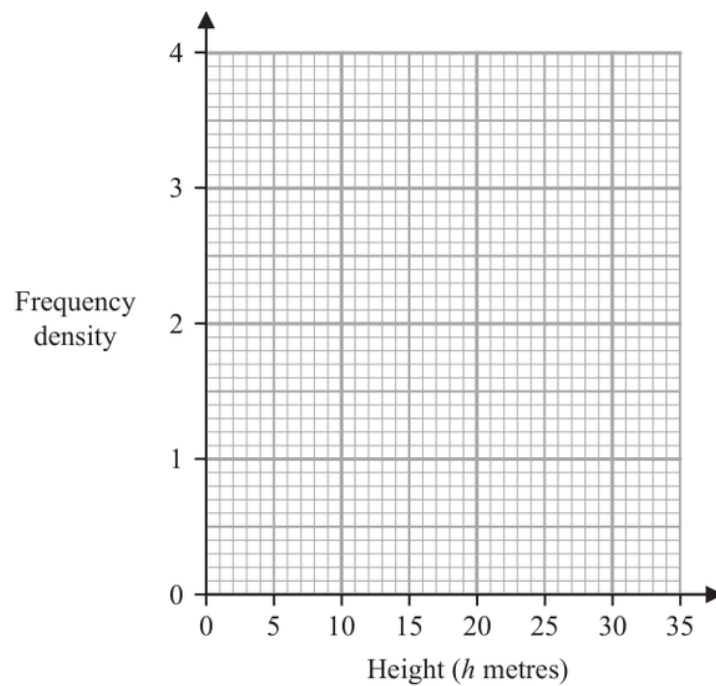
PAST PAPER QUESTION**Q#: 16****Marks: 3****TOPIC: Histograms**

May 2024 P1HR - May 2024 | P1HR

16 The table gives information about the heights, in metres, of the trees in a park.

Height (h metres)	Frequency
$0 < h \leq 2$	5
$2 < h \leq 5$	12
$5 < h \leq 10$	18
$10 < h \leq 20$	14
$20 < h \leq 35$	9

On the grid, draw a histogram for this information.



(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 3

TOPIC: **Histograms**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2024 P1HR - May 2024 | P1HR

17 (a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

$$n = \dots\dots\dots$$

(1)

(b) Express $\frac{7}{2-\sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

$$\dots\dots\dots$$

(3)

(Total for Question 17 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 17

Marks: 4

TOPIC: **Algebraic Roots & Indices**

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WORKED SOLUTION

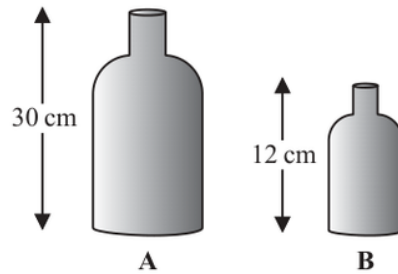
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

May 2024 P1HR - May 2024 | P1HR

18 The diagram shows two similar vases, **A** and **B**Diagram **NOT**
accurately drawnThe height of vase **A** is 30 cmThe height of vase **B** is 12 cm

Given that

$$\text{surface area of vase A} - \text{surface area of vase B} = 178.5 \text{ cm}^2$$

find the surface area of vase **A**..... cm²**(Total for Question 18 is 4 marks)**

PAST PAPER SOLUTION

Q#: 18

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Differentiation**

May 2024 P1HR - May 2024 | P1HR

19 A curve C has equation $y = x^3 - 8x^2 - 12x + 5$

Curve C has exactly two stationary points, one at point A and one at point B such that

x coordinate of point $A > x$ coordinate of point B

Find the coordinates of point A

Show clear algebraic working.

(.....,)

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 19

Marks: 5

TOPIC: **Differentiation**

May 2024 P1HR - May 2024 | P1HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Completing the Square**

May 2024 P1HR - May 2024 | P1HR

- 20 (a)** Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a , b and c are numbers to be found.

.....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

- (b) Find the coordinates of P

(..... ,)

(2)

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Completing the Square**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Completing the Square**

May 2024 P1HR - May 2024 | P1HR

- 20** (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a , b and c are numbers to be found.

.....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

- (b) Find the coordinates of P

(..... ,)

(2)

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Completing the Square**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

21 There are 25 counters in a bag such that

6 counters are blue

x counters are orange, where $x > 9$

the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Probability Toolkit**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

21 There are 25 counters in a bag such that

6 counters are blue
 x counters are orange, where $x > 9$
the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.
Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Probability Toolkit**

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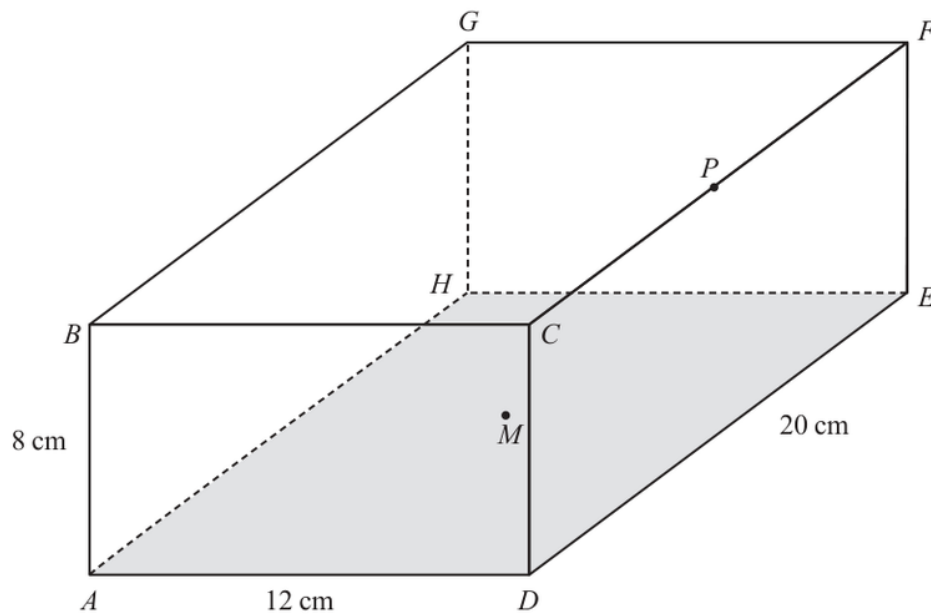
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: 3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

22 The diagram shows a cuboid $ABCDEFGH$ with horizontal base $ADEH$ Diagram **NOT**
accurately drawn

$AB = 8 \text{ cm}$

$AD = 12 \text{ cm}$

$DE = 20 \text{ cm}$

 M is the midpoint of the base $ADEH$ and P is the midpoint of the edge CF Work out the size of angle BMP

Give your answer correct to one decimal place.

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

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PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

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WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Sequences**

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23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 4

TOPIC: **Sequences**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Sequences**

May 2024 P1HR - May 2024 | P1HR

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 4

TOPIC: **Sequences**

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